

Renoir Boulanger

Expert in Front-End Ops, UI Libraries who crafts accessible web experiences with precision

SUMMARY

Experienced full-stack developer dedicated to creating framework-agnostic web solutions that excel in Accessibility, Performance, and Code Quality. Strong emphasis on Web Standards, continuous testing, and test-driven development (TDD). A polyglot programmer contributing to open-source projects and architecting robust package management systems. Designed and maintained hosting infrastructure for high-visibility sites with expertise in Cloud IaC, Containers, and continuous deployment systems focusing on reproducible builds and consistent environments.

- Polyglot programmer implementing precise solutions in Deno, TypeScript, ECMAScript, JavaScript, PHP, Bash, GNU Make, C#, Python
- Expertise in Web Standards, Accessibility (WCAG), Internationalization (i18n), and Localization (l10n)
- Experience in Infrastructure As Code with Salt Stack, Vagrant, Ansible, Puppet
- Technical contributor to open-source projects with publicly available implementations: Nuxt.js documentation, Alpine Linux package maintenance, MediaWiki compatibility tables, and Mozilla Firefox Accounts
- Developed and documented reusable technical solutions like type-safe bitmask operations in JavaScript, Salt Stack infrastructure states, and automated MediaWiki content conversion tools
- Architected documented hosting infrastructure for high-visibility sites including W3C's WebPlatform Docs project <https://webplatform.github.io/> and The World Wide Web's 25th anniversary site <https://www.w3.org/webat25/>
- Deep expertise in Cloud IaaS, Containers, and continuous deployment systems with emphasis on reproducible builds and consistent environments

EXPERIENCE

Self-Employed — *Open Source Web Components Specialist (part time)*

September 2024 - Present

Developing framework-agnostic "headless UI" web component libraries using native browser capabilities and Web Standards. Implementing modern serverless architectures with Deno and CloudFlare Workers for edge computing. Creating a modular system that implements W3C Web Components Community Group "Context" community protocol to enable seamless integration across any JavaScript framework environment.

- Creating and documenting implementations of the W3C Web Components Community Group's Context API protocol, enabling sophisticated component communication without direct dependencies
- Architected renoirb-esm-modules (<https://github.com/renoirb/renoirb-esm-modules/>): A collection of unstyled, framework-agnostic Web Components that consume structured data to generate consistent, accessible experiences
- Developed a data-driven document generation system where a single YAML source produces both PDF documents and interactive web experiences — or this document —, demonstrating clear separation of content and presentation.
- Created Archivator (npm: archivator): A system for content preservation enabling historical comparison, indexing, and offline search capabilities (<https://renoirb.github.io/archivator-demo/>)

Montréal, Quebec

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<https://renoirboulanger.com>

<https://renoirb.com/resume/rework.html>

<https://linkedin.com/in/renoirb>

<https://github.com/renoirb>

<https://gitlab.com/renoirb>

SKILLS

Programming

PHP, C#, VCL, Bash, Python

Web Development Languages

TypeScript, JavaScript, ECMAScript, HTML/CSS

PHP Frameworks

Symfony, Slim, Composer/Packagist

Frontend Frameworks

Nuxt.js, Vue.js, Koa.js, Element UI, Backbone.js, Marionette.js, Angular

Web Development Libraries

Rollup, Bili, Rush.js

Frontend Testing Frameworks

Jest, CodeceptJS, Playwright, AVA, Mocha, Jasmine

Web Platform

Accessibility, Adaptive, Bundling, Deployment, HTML/CSS Architecture, Polyfills, Promises, Responsive, Transpilation

Web Operations

Configuration Management, Docker Swarm, Dockerflow, Elasticsearch, Node.js, Prometheus, Redis, Salt Stack, Self-Healing event handlers, Service Monitoring, Varnish

EDUCATION

Université de Sherbrooke, CeFTI

INF731 - Object-Oriented

- Created tuples-boolean-bitmasker (npm: tuples-boolean-bitmasker): A library that converts object boolean properties into bitmasks for efficient, type-safe testing of complex conditional states in JavaScript applications

Full-time parenting — *Father*

June 2022 - September 2024

Full-time caregiver for children.

GoTo Inc. — *Staff Software Developer (remote)*

January 2021 - May 2022

Contributed to an established micro-frontend architecture based on Web Components and lit-element, helping teams leverage its framework-agnostic foundations. Provided technical guidance on native Web Platform capabilities, demonstrating how to maintain accessibility and performance while integrating with React, and other frameworks. Created documentation and working examples showing how to effectively use the existing component system across different technology stacks.

- Contributed architectural improvements to the Application Shell infrastructure, enabling seamless integration of diverse product experiences
- Helped development teams understand and effectively utilize the Web Components architecture, ensuring consistent implementation across different framework contexts
- Built and documented integration patterns demonstrating how to properly implement new features within the established component system
- Developed examples and test suites validating accessibility and internationalization implementation within the architecture

RBC Royal Bank of Canada Inc. — *Senior Full Stack Developer (remote)*

March 2020 - December 2020

At RBC, I served as a Senior Full-Stack Developer, leading the implementation of scalable modules for the RBC Omni platform (Online Business Banking). I collaborated closely with cross-functional teams to deliver solutions aligned with business objectives. Noteworthy achievements include streamlining developer onboarding processes and enhancing legacy systems for improved performance and reliability.

- Make implementation proposals driven by the requirements and agenda set by product owners and business analysts.
- Build scalable and reusable modules by collaborating with other production teams, including the design system/platform team.
- Rewritten new developer onboarding documentation and completely automated how to properly setup the current password to be used properly through Git and other scripting, including the Bank's authoritative self-signed certificate that would cause many headaches.

Université de Sherbrooke (cntd.)

Programming in C#

INF743 - Software Architecture

INF747 - Designing and managing Enterprise Information Systems

June 2024

Udemy

Nuxt.js - Vue.js on Steroids

Understanding TypeScript

EggHead.io

Use Types Effectively in TypeScript

Up and Running with TypeScript

Get Started with Elasticsearch

Build a Server Rendered Vue.js

App with Nuxt and Vuex

VueMastery

Real World Vue.js

Mastering Vuex

Next-Level Vue

Scaling Vue with Nuxt.js

Advanced Components

Pearson Vue

Zend Certified Engineer PHP 5.3

July 2013

McGill University

CEEN-421-761 - English in Written Communication Contexts

CMSC-000-781 - Foundations of Mathematics

CMSC-101-771 - College Algebra and Functions

December 2013

University of Montreal

Web Accessibility (WCAG 1) at the Nazareth & Louis Braille institute

July 2008

IBM

IBM WebSphere DataPower SOA Appliance

July 2007

CGI Inc. — Senior Technical Consultant (Lead Frontend) (hybrid)

May 2017 - February 2020

As a Senior Technical Consultant, I led the development of Unify360, a Multi-Cloud Management Platform. Leveraging microservices architecture and atomic design principles, I orchestrated the design and implementation of a robust frontend solution deployed across multiple regions. My role involved extensive experimentation with modern technologies and mentoring team members to foster a culture of innovation.

- Experiment and document Frontend software development platform technological choices
- Lead the team that work and coach for writing code with tests, making self-contained and re-usable packages

AlayaCare — Senior Software Developer

November 2016 - May 2017

During my tenure at AlayaCare, I focused on optimizing Docker Compose builds and enhancing team productivity through streamlined workspace setups. I led initiatives to automate development environment configurations, reducing setup time from days to hours. My efforts significantly improved collaboration and efficiency across the software development team.

- Overhaul of the Software development team's workspace setup, which allowed anyone to have exactly the same setup by invoking a few simple commands. A fresh computer would be ready after 1h instead of taking multiple days and asking around to other colleagues.

Betastream — DevOps Automation Specialist (part time, remote)

January 2016 - June 2016

Designed and implemented a comprehensive infrastructure architecture to replace manual AWS deployments, establishing a vendor-neutral cloud orchestration system with standardized deployment patterns across environments. Created a security-focused approach with clear separation between public-facing and administrative interfaces, mandatory TLS encryption, and consistent naming conventions across all system components.

- Architected complete cloud orchestration solution using SaltStack that worked consistently across AWS, Google Cloud, and Azure, with standardized naming conventions and deployment paths
- Established comprehensive security model separating public origins from admin interfaces with distinct hostnames, IP restrictions, and client certificate authentication
- Implemented atomic deployment system using timestamped archives and symbolic links, enabling zero-downtime deployments and instant rollbacks across all application servers
- Integrated Fastly CDN with custom Varnish Configuration Language (VCL) testing environment and consistent TLS implementation across all communication layers

Canadian Armed Forces

Basic training at Royal Military

College Saint-Jean

Private (Basic), assigned to

Sherbrooke 714th Communication Squadron

Wired and wireless

communication and information

systems (Introductory training)

Winter warfare training

January 2000

PUBLICATIONS

- Santachè, A. "Mobile development using web technologies focusing on games", ACM, ISBN 978-1-4503-2559-2
Participated in chapters related to "Markup architecture", "Events" and "Web Storage"
- Kumar, J. "Apache Solr PHP Integration", Pakt Publishing, ISBN 9781782164920
Book review
- « Techniques pour déployer WordPress de façon consistante et les rendre résilients », WordCamp Montréal 2016
Talk given at WordCamp Montréal 2016
- "Packaging for easier redeployment", DevOps Montréal
Presented at two events, DevOps Montréal, PHP Québec.
- « Comment évaluer le niveau de qualité d'un site web selon les techniques d'intégration web d'actualité. », Web à Québec 2013
Presented at two conferences; « Montreal WebIn », and « Web à Québec » (WAQ)

CAE — *DevOps Automation Specialist (part time)*

March 2016 - June 2016

Deliberately brought in as a "disruptor" to introduce modern Infrastructure as Code (IaC) practices to CAE's established environment. While another department used Chef (Ruby-based), I implemented accessible solutions using Bash scripts - a language the existing team was already comfortable with. I developed deployment and monitoring solutions for air-gapped networks with self-managed TLS certificate authorities, establishing patterns for securely automating systems that were previously manually configured, while meeting strict aerospace industry security requirements.

- Collaboratively introduced Infrastructure as Code (IaC) principles through accessible technologies (Bash), meeting the team where they were comfortable rather than imposing unfamiliar Ruby-based solutions
- Created automation scripts that configured developer tools (Git, Python, NPM, cURL) to explicitly trust internal certificate chains, solving a common pain point in secured environments
- Developed deployment techniques for "playbooks" packages that could function within airtight network boundaries while maintaining consistency across environments
- Evaluated and implemented monitoring solutions including Prometheus, Mozilla Heka, Monit, and Grafana, adapting them to function with corporate TLS certificates

Mozilla — *Subject Matter Expert Compatibility Data Project*

August 2015 - January 2016

Contracted as a Product Owner for the Browser Compatibility Data project based on my previous experience with W3C's WebPlatform project. My role primarily involved participating in discussions regarding identifier systems for browser compatibility data, attending meetings, and preparing documentation and slide decks to communicate technical concepts to stakeholders.

- Communicated with various stakeholders about the requirements and features for the compatibility data system
- Participated in discussions about identifier schema designs for browser compatibility data
- Created presentation materials and documentation explaining technical concepts
- Attended meetings to represent the project's technical needs

W3C/MIT — *Developer Operations Engineer*

August 2013 - August 2015

At W3C/MIT, I played a key role in maintaining and enhancing the WebPlatform Docs project, overseeing infrastructure stability and feature development. I implemented robust monitoring solutions and streamlined deployment processes, ensuring optimal performance and reliability. My efforts contributed to the project's success in becoming a premier resource for web developers worldwide.

- Convert and preserve many hundred thousand edits, and keep trace of the content edition contributors and date of the change from MediaWiki into Static site generators and Git (2015)
- Improved Infrastructure deployment code so that any service could be rebuilt systematically (2015)
- Designed and Implemented Single Sign-On solution using OAuth2 (2014)

Ericsson — *Web Developer*

February 2012 - June 2013

As a Web Developer at Ericsson, I led the development of a web-based virtualization provisioning portal, enabling users to initiate virtual machines across geographically dispersed data centers. My role involved architecting and implementing innovative solutions using cutting-edge technologies, driving improvements in scalability and performance.

- Architect, build and lead a Web-based “Virtualization provisioning” portal.

Evocatio Solutions technologiques Inc. — *Senior Web Developer, Co-Founder*

January 2010 - March 2012

As Senior Web Developer and Co-Founder of Evocatio Solutions, I led web development projects for prestigious clients, including UDA and Astral Media. My responsibilities encompassed architecting frontend solutions, collaborating with graphic designers, and optimizing web applications for performance and usability. Notable achievements include spearheading major website rewrites and implementing advanced e-mail delivery systems.

- Lead and architect all Web Development work related to the FrontEnd layer of the stack.
- In touch with Graphic Designers and Project Managers to create valid HTML/CSS markup that reflects the designs, and to work on all Web Browsers of the moment.
- Take Adobe Photoshop and PDFs document and create HTML/CSS Static style guides (minimal JavaScript, no back-end) to create CSS pattern libraries.

Groupe Informatique TechSolCom Inc. — *Web Developer*

October 2006 - August 2009

During my tenure at Groupe Informatique TechSolCom Inc., I played a pivotal role in maintaining the company's web presence and contributing to internal R&D projects. I led the front-end development aspect of a proprietary CI server web application and participated in building inventory management systems. My contributions enhanced operational efficiency and scalability across various projects.

- Maintain the company corporate image on the web, maintain their Web site, and participate in projects run internally as a Web developer on R&D projects.
- Participated in maintaining company internal network, testing DataPower blade, Virtualized servers on bare metal, and web hosting infrastructure.
- Participated in building a proprietary CI server Web Application (“Beebox”) for IBM WebSphere that was intended to be a competitor to Jenkins and Cruise Control of the time. (2007-2009)
- Contributed to Inventory management application (“Namminik”) as an internal R&D experiments to make a Web Application also have its native iOS client. (2009)

Câble Axion Digital — *Web Developer*

October 2005 - November 2006

My role was to rewrite the corporate website, maintain internal applications and answer to level-2 support calls from subscribers.

INEXIS Solution web Inc. — *Web Developer, Owner*

February 2002 - November 2006

INEXIS Solution web Inc., was a one-man web development service provider firm I founded to serve graphic design agencies in the region of the Eastern Townships. Services included managing own private Web hosting infrastructure and building HTML/CSS pages based on PDF files provided by graphic designers, creating e-mail templates.